Contextual Predictability in Negation is not Special: General Facilitation but no Structural Effects

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Introduction

Predictability and negation processing

- A classic interaction in negation processing: false affirmatives are costlier (longer RTs, larger N400) than true affirmatives, but **true negatives** are costlier than false negatives[1]

True, Affirmative (TA)	A robin is a bird.	
False, Affirmative (FA)	A robin is a tree.	→ **+cost**
True, Negative (TN)	A robin is not a tree.	→ **+cost**
False, Negative (FN)	A robin is not a bird.	

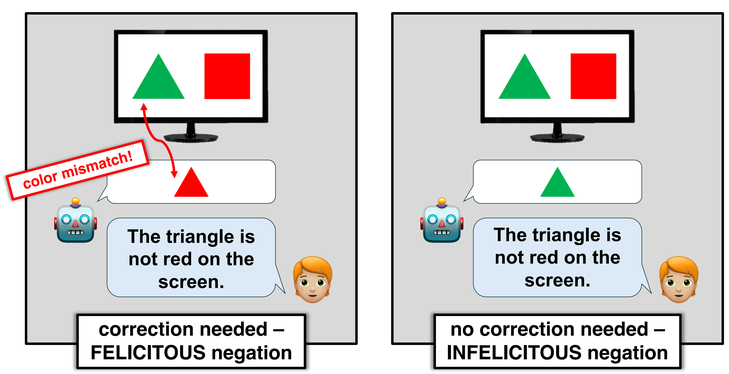
Research question

Does prediction exert a general facilitatory role [2-3] or a deeper influence[4] on negation processing?

- Requires accounting for the pragmatics of negation: the polarity-truth value interaction is notoriously modulated by this factor[5-7]

Approach

An incompetent robot being trained to recognize geometric shapes invited **corrections**[8] from time to time:



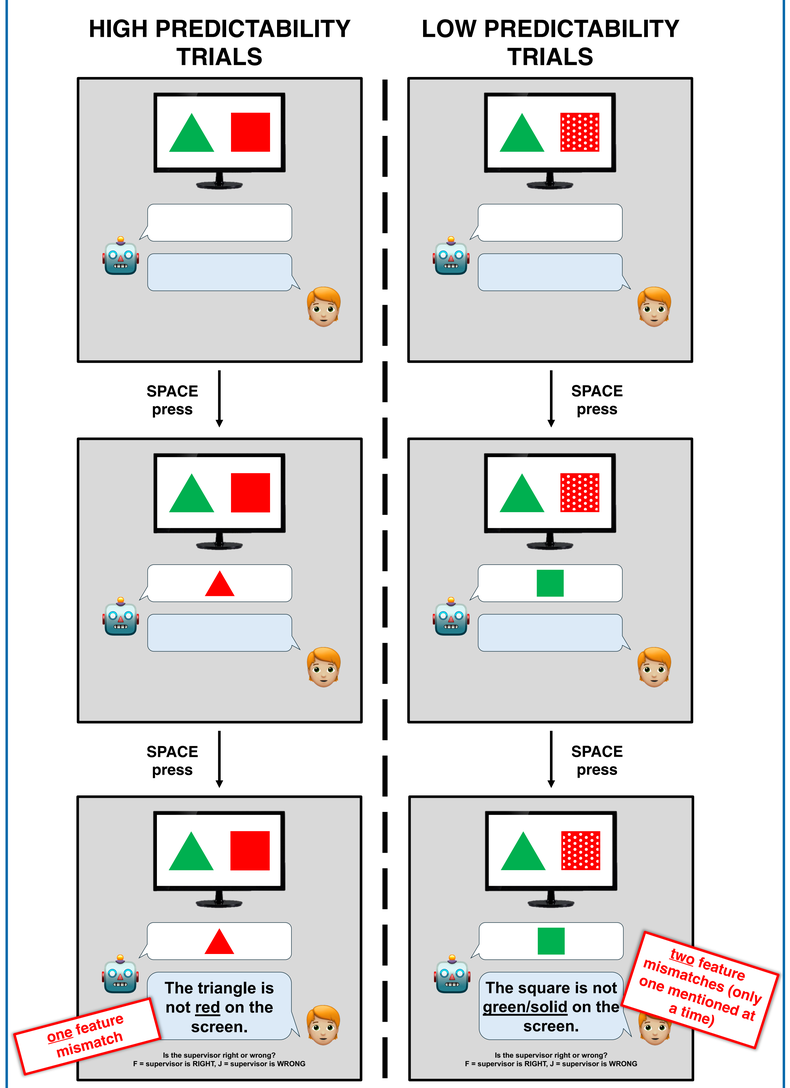
Methods

Sentence verification task (80 items, N = 75)

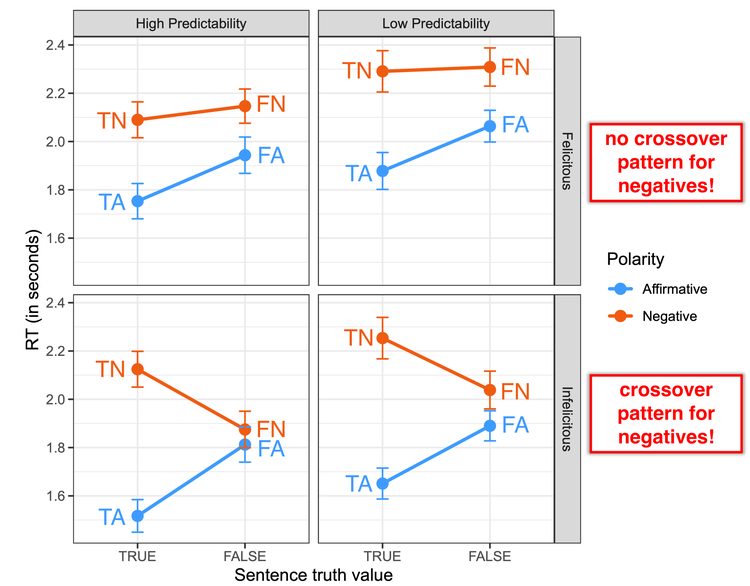
Independent variables (within-subjects):

- Predictability level (high/low)
- Pragmatic context (felicitous/infelicitous)
- Sentence polarity (affirmative/negative)
- Sentence truth value (true/false)

Dependent variable: judgment RT



Results



- Successful replication of the **three-way interaction between pragmatic context, polarity, and truth value** ($\chi^2(1) = 38.9$, $p < .001$; see top vs. bottom panels above)
- Robust **main effect of predictability** ($\beta = -0.04$, SE = .005, $t = -7.00$, $p < .001$; see left vs. right panels above), **but (very) strong evidence against its involvement in higher-order interactions** ($BF_{10} = 0.026$)

Takeaway/TL;DR

Contextual predictability has a **general facilitatory role** rather than a deep involvement in negation processing, even when we account for the pragmatic requirements of negation use.

References and Acknowledgments

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